Supplementary Material: Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

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APPENDIX A
COMPLETE PER-SEED BENCHMARK EVALUATION

To ensure thorough scientific transparency and enable exact experimental reproducibility, this section documents all individual seed runs across the four factorial configurations on the standard Flickr8k held-out test set (1,000 images, 5,000 captions). Table I details the complete directional recall metrics (Image-to-Text and Text-to-Image at top-1, top-5, and top-10) alongside Validation and Test Mean Recall. Table II summarizes the corresponding training wall-clock time and peak GPU memory allocation.

APPENDIX B
HARDWARE AND EXECUTION ENVIRONMENT PROVENANCE

All 12 training and evaluation runs were executed on a dedicated cloud compute instance under identical software and hardware specifications:

- **Accelerator Hardware:** NVIDIA Tesla T4 GPU (16 GB GDDR6 VRAM, Compute Capability 7.5).
- **Host System:** Linux / Windows Server x86_64 multi-core platform.
- **Software Stack:** PyTorch 2.1.0+cu121, Torchvision 0.16.0+cu121, CUDA 12.2, Driver 535.104.05.
- **Pretrained Model Weights:** `timm` Vision Transformer (`vit_base_patch16_224`, 86.6 M parameters, frozen), HuggingFace Transformers `roberta-base` (124.6 M parameters, frozen).
- **Determinism Protocol:** CUDA deterministic algorithms enabled, random seed explicitly set for Python random, NumPy, and PyTorch (CPU and CUDA).

APPENDIX C
ADDITIONAL IMPLEMENTATION DETAILS AND REPRODUCIBILITY

- **Optimization Routine:** AdamW optimizer with $\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}, initial learning rate $\eta_{\text{max}} = 2 \times 10^{-4}$, cosine decay schedule down to $\eta_{\text{min}} = 10^{-6}$ over 8 epochs.
- **Gradient Management:** Global gradient norm clipping at 1.0 applied before each optimizer update step.
- **Batch Formation:** 64 images per batch paired with 5 captions each, yielding an effective batch size of 320 image-caption pairs.
- **Model Checkpointing:** Validation Mean Recall evaluated at the conclusion of every epoch; the highest-performing checkpoint is selected for final evaluation on the certified held-out test split.

TABLE I

COMPLETE 12-RUN PER-SEED TEST RETRIEVAL METRICS AND VALIDATION MEAN RECALL ON THE STANDARD FLICKR8K TEST SPLIT. I2T AND T2I DENOTE IMAGE-TO-TEXT AND TEXT-TO-IMAGE RETRIEVAL, RESPECTIVELY; ALL VALUES ARE REPORTED AS PERCENTAGES (%).

Model Configuration	Seed	Best Ep.	Val MR	I2T@1	I2T@5	I2T@10	T2I@1	T2I@5	T2I@10	Test MR
Custom SSD-Style Baseline	42	8	48.03	27.00	57.60	70.10	21.24	50.16	65.18	48.55
	43	8	48.01	26.30	56.20	68.60	19.94	49.78	64.68	47.58
	44	8	47.78	26.30	58.10	71.60	20.18	49.74	65.74	48.61
HEDO-HVSC w/o HEDO	42	8	48.08	27.70	57.90	69.30	20.52	50.14	65.02	48.43
	43	8	48.14	25.70	55.90	69.20	19.78	50.04	65.08	47.62
	44	8	47.67	26.70	57.40	71.60	20.58	49.66	65.72	48.61
HEDO-HVSC w/o HVSC	42	8	48.03	27.90	56.70	69.80	21.28	50.08	64.84	48.43
	43	8	49.16	25.80	57.10	70.20	19.98	50.28	65.04	48.07
	44	8	48.70	27.90	57.10	70.90	20.72	50.62	65.42	48.78
Full HEDO-HVSC (Ours)	42	8	47.86	28.20	56.50	69.90	20.88	50.14	64.84	48.41
	43	7	48.56	24.10	56.50	69.90	20.46	49.42	64.62	47.50
	44	8	48.60	27.90	57.50	71.60	20.78	50.36	65.42	48.93

TABLE II

PER-SEED TRAINING WALL-CLOCK TIME AND PEAK GPU VRAM FOOTPRINT ACROSS ALL 12 BENCHMARK RUNS.

Model Configuration	Seed	Best Ep.	Train Time	Peak VRAM
Custom SSD-Style Baseline	42	8	17.51 min	1875.76 MB
	43	8	17.47 min	1878.22 MB
	44	8	17.47 min	1877.78 MB
HEDO-HVSC w/o HEDO	42	8	17.72 min	1878.61 MB
	43	8	17.84 min	1878.37 MB
	44	8	17.90 min	1878.81 MB
HEDO-HVSC w/o HVSC	42	8	17.56 min	1878.99 MB
	43	8	17.56 min	1879.74 MB
	44	8	17.60 min	2685.66 MB
Full HEDO-HVSC (Ours)	42	8	17.81 min	2686.05 MB
	43	7	17.80 min	2686.25 MB
	44	8	17.85 min	2686.25 MB